FA 550: Data Visualization

Capstone Final Project Documentation

CORD-19 Research Trends Dashboard

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Tableau Public Link: https://public.tableau.com/app/profile/manoj.kumar.ilango/viz/Book1_projectcapstone2026/CORD-19researchTrendsDashboard

Section 1: Executive Summary

This project presents the CORD-19 Research Trends Dashboard, an interactive executive dashboard built using Tableau Desktop that visualizes how COVID-19 scientific research evolved from 1990 to 2022. The dashboard analyzes the COVID-19 Open Research Dataset Challenge (CORD-19), a large publicly available collection of scholarly articles on COVID-19 and related coronaviruses published by the Allen Institute for AI.

Tableau Public Link: https://public.tableau.com/app/profile/manoj.kumar.ilango/viz/Book1_projectcapstone2026/CORD-19researchTrendsDashboard

Presentation Video: https://www.youtube.com/watch?v=OHEbeO0iaQQ

Key Features

- KPI Summary Cards: Four high-level metrics showing 499,714 total unique papers, 19,401 journals, 48 source collections, and 1,259 average papers per month
- Publication Trend Line Chart: Monthly research output from 1990 to 2022 with average reference line highlighting the COVID-19 research explosion
- Research Output Heatmap: Month-by-year grid encoding research intensity through color, clearly revealing the 2020-2022 COVID peak
- Top Journals Bar Chart: Horizontal bar chart showing BioRxiv, PLOS ONE, and Int J Environ Res Public Health as dominant publishers
- Source Collections Bar Chart: Distribution of papers across Medline, PMC, WHO, BioRxiv, and other repositories
- Journal Trends Multi-line Chart: Comparative growth trajectories of top 15 journals from 2005 to 2022
- Yearly Volume Bar Chart: Annual paper counts clearly showing the 2021 historic peak of 200,000+ papers
- Keywords Treemap: Research theme frequency visualization extracted from paper titles

Technologies Used

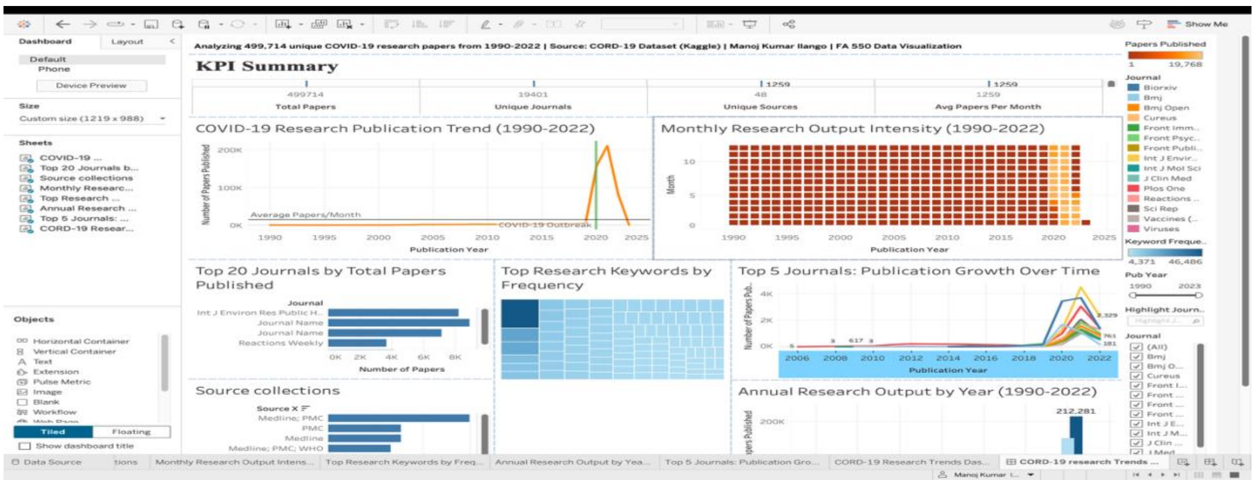
- Python 3 with pandas' library for data cleaning and transformation
- Tableau Desktop for dashboard design and visualization
- Tableau Public for deployment and sharing
- CORD-19 dataset from Kaggle (Allen Institute for AI)

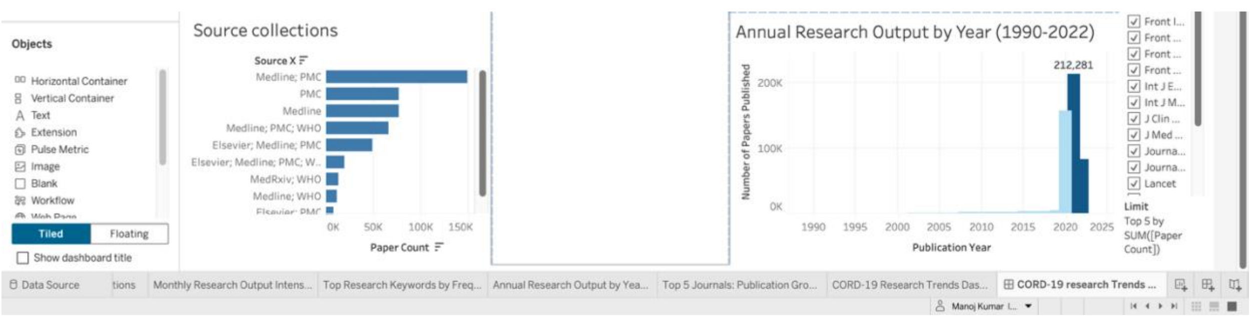
Key Findings

- COVID-19 research output increased by over 2,000% between 2019 and 2021, peaking in March 2021 with nearly 20,000 papers published in a single month
- BioRxiv emerged as the dominant publisher with approximately 8,800 papers, reflecting the pandemic-era shift toward preprint publishing for faster knowledge sharing
- Research themes shifted dramatically toward clinical topics including treatment, vaccine, respiratory, and care during 2020-2021

Impact and Value

This dashboard provides researchers, policymakers, students, and data analysts with an intuitive tool to explore how the global scientific community responded to COVID-19. By transforming over 499,000 raw metadata records into interactive visual insights, the dashboard reveals publication patterns, journal dominance, and research theme evolution that would be impossible to discern from raw data alone.





Section 2: Project Goals and Context

Problem Statement

The COVID-19 pandemic generated an unprecedented volume of scientific literature. Within the first two years of the pandemic, more research papers were published on COVID-19 than on any other topic in the history of modern medicine. However, this volume of information created its own challenge: how can researchers, administrators, and policymakers quickly understand the landscape of COVID-19 research without reading thousands of papers?

This project addresses that challenge by creating an executive dashboard that transforms raw publication metadata into actionable visual insights. The dashboard answers key questions such as: When did COVID-19 research peak? Which journals led the response? How did research themes evolve over time?

Project Objectives

- Build a fully interactive Tableau executive dashboard using the CORD-19 dataset with at least 8 connected visualizations
- Enable users to filter and explore publication data by time period journal, and source collection

- Reveal meaningful insights about COVID-19 research publication patterns without requiring technical expertise from users
- Demonstrate professional data visualization skills including data cleaning, pipeline development, and dashboard design
- Produce a portfolio-quality project suitable for presentation to potential employers

Target Audience

The dashboard is designed for three primary audiences. First, researchers and academics who want to understand the landscape of COVID-19 literature and identify leading journals in their field. Second, data analysts and visualization professionals who can use this project as a reference for working with large metadata datasets. Third, entry-level recruiter audiences and hiring managers in data analytics who are evaluating portfolio projects. The dashboard uses clear, non-technical language throughout and avoids jargon so that users without a research background can navigate it comfortably. All chart titles, axis labels, and tooltips are written in plain English.

Personal Motivation

I chose this project because it sits at the intersection of public health, data analytics, and information visualization — three fields I find compelling. The CORD-19 dataset offers a rare opportunity to work with real-world, large-scale academic metadata while exploring a topic of genuine global significance. Building an executive dashboard around COVID-19 research also allows me to demonstrate skills directly relevant to roles in healthcare analytics, research intelligence, and business intelligence.

Section 3: Data Preparation Process

Data Source

The primary data source is the CORD-19 (COVID-19 Open Research Dataset Challenge) dataset, publicly available through Kaggle at https://www.kaggle.com/datasets/allen-institute-for-ai/CORD-19-research-challenge. The dataset was compiled by the Allen Institute for AI in partnership with leading research groups and contains machine-readable scholarly articles on COVID-19 and related coronaviruses.

Metric	Value
Primary file	metadata.csv
Original file size	~757 MB (uncompressed)
Original row count	1,056,660 records
Final cleaned row count	499,714 unique papers
Date range	1990 – 2022 (COVID peak: 2020–2021)
Key fields used	cord_uid, title, abstract, publish_time, authors, journal, doi, source_x
Source collections	PubMed, PMC, WHO, BioRxiv, MedRxiv, Elsevier, ArXiv

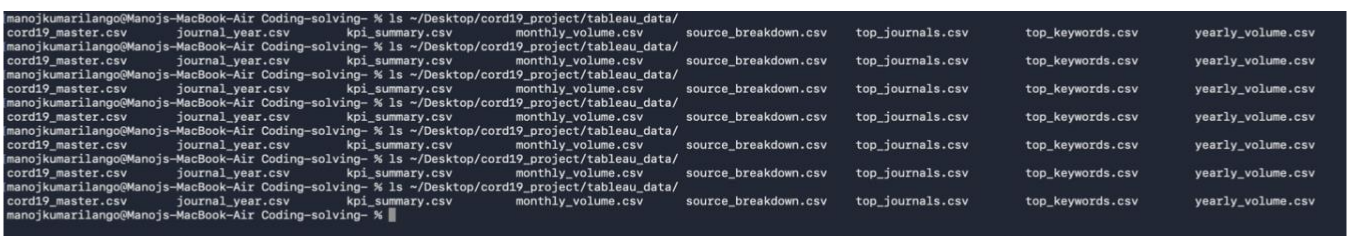
Data Quality Assessment

Upon initial inspection of the raw metadata file, several data quality issues were identified. The dataset contained significant duplication because the same paper was often indexed by multiple source collections simultaneously. For example, a paper published in a PMC journal might appear under both the PMC and WHO source collections, resulting in two or more identical rows for the same research article. Additionally, the publish time field used inconsistent date formats across records, with some entries containing full dates (YYYY-MM-DD), others containing only years (YYYY), and some containing invalid or missing values. Journal names showed inconsistent capitalization and spacing, and approximately 8% of records had blank journal fields.

Cleaning and Transformation Pipeline

All data cleaning was performed using Python 3 and the pandas library. The following steps were executed in sequence:

- Step 1 – Column Selection: Only the 8 required fields were loaded from the 757MB file to reduce memory usage. This reduced the working dataset size from approximately 757MB to under 200MB in memory.
- Step 2 – Deduplication by cord_uid: Each paper in CORD-19 has a unique cord_uid identifier. Duplicate rows sharing the same cord_uid were removed, keeping the first occurrence. This deduplication was essential because the CORD-19 dataset intentionally indexes the same paper across multiple source collections for completeness. Without deduplication, any analysis of publication volume would be inflated by 53%, misrepresenting the true volume of unique research
- Step 3 – Deduplication by DOI: A secondary deduplication pass using the doi field removed any remaining cross-source duplicates not caught by cord_uid matching. Together these two steps reduced 1,056,660 rows to 499,714 unique papers.
- Step 4 – Date Parsing and Filtering: The publish_time field was parsed to datetime format using pandas to_datetime with errors set to coerce, which converted unparseable values to NaN. Records outside the range of 1990 to 2023 were filtered out as invalid or out of scope.
- Step 5 – Derived Date Fields: Three new date fields were created: pub_year (integer year), pub_month (integer month 1-12), and year_month (YYYY-MM string for use as the Tableau time axis).
- Step 6 – Journal Name Standardization: Journal names were stripped of leading and trailing whitespace, converted to title case for consistency, and blank values were replaced with the string Unknown.
- Step 7 – Source Collection Cleaning: The source_x field was stripped of whitespace and null values were replaced with Unknown.
- Step 8 – Author Count Derivation: A new author_count field was calculated by counting semicolon-delimited entries in the authors field, providing a proxy measure of collaboration intensity per paper.
- Step 9 – Summary Table Export: Eight purpose-built CSV files were exported to a tableau_data folder for direct connection to Tableau Desktop.



Output Files

File	Description	Rows
cord19_master.csv	Full cleaned dataset with all key fields	499,714
monthly_volume.csv	Paper count by year, month, and year-month	397
yearly_volume.csv	Paper count aggregated by year	34
top_journals.csv	Top 20 journals by paper count	20
source_breakdown.csv	Paper count by source collection	48

journal_year.csv	Top 15 journals by paper count per year	147
kpi_summary.csv	Single-row KPI metrics table	1
top_keywords.csv	Top 100 keywords from paper titles by frequency	100

Section 4: Design Decisions and Rationale

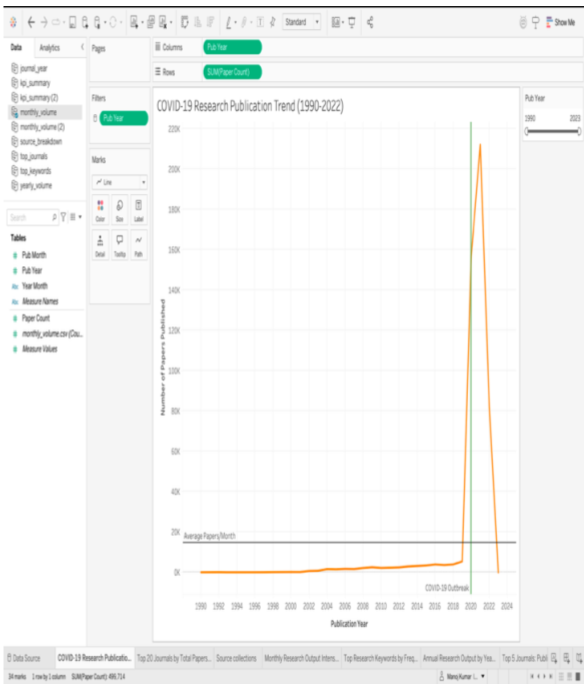
Dashboard Layout and Visual Hierarchy

The dashboard follows a standard executive dashboard layout with KPI cards at the top and detailed charts below. This layout is well-established in business intelligence design because it allows users to immediately see high-level summary metrics before diving into details. The top-to-bottom flow mirrors how users naturally scan screens and prioritize the most important numbers (total papers, journals, sources, average output) prominently. The dashboard is organized into three visual rows: KPI summary at the top, time-series and heatmap in the middle row for temporal analysis, and journal, keyword, source, and yearly charts in the bottom row for categorical analysis. This separation ensures that temporal and categorical insights are visually grouped and easy to compare.

Visualization Selection Rationale

Chart	Type Chosen	Alternatives Considered	Reason for Choice
Publication Volume	Line Chart	Area chart, bar chart	Line charts are optimal for continuous time series data showing trends. The smooth line clearly reveals the sudden 2020 inflection point better than bars would.
Research Intensity	Heatmap	Stacked bar, bubble chart	The month x year grid format perfectly encodes two temporal dimensions simultaneously. Color intensity immediately draws attention to the 2020-2022 peak months.
Top Journals	Horizontal Bar Chart	Vertical bar, treemap	Horizontal bars are preferred when category labels (journal names) are long text strings. They allow full journal names to be readable without rotation.
Keywords	Treemap	Word cloud, bubble chart	Tableau does not natively support word clouds. Treemaps encode frequency through area size and color intensity simultaneously, making them superior to bubble charts for ranked categorical data.
Journal Trends	Multi-line Chart	Small multiples, stacked area	Multiple lines on one chart enables direct visual comparison of journal growth trajectories, which is the primary insight goal of this view.

Yearly Volume	Bar Chart	Line chart, area chart	Discrete yearly totals are more accurately represented by bars than by a continuous line, which could imply interpolated values between years.
Source Collections	Horizontal Bar	Pie chart, treemap	Pie charts fail beyond 5-6 categories. With 48 source collections, a sorted horizontal bar chart enables accurate length comparison and ranking.
KPI Summary	Text Table	Big number tiles, gauge charts	A clean text table with large numbers communicates KPI values directly without visual embellishment, appropriate for an executive audience.



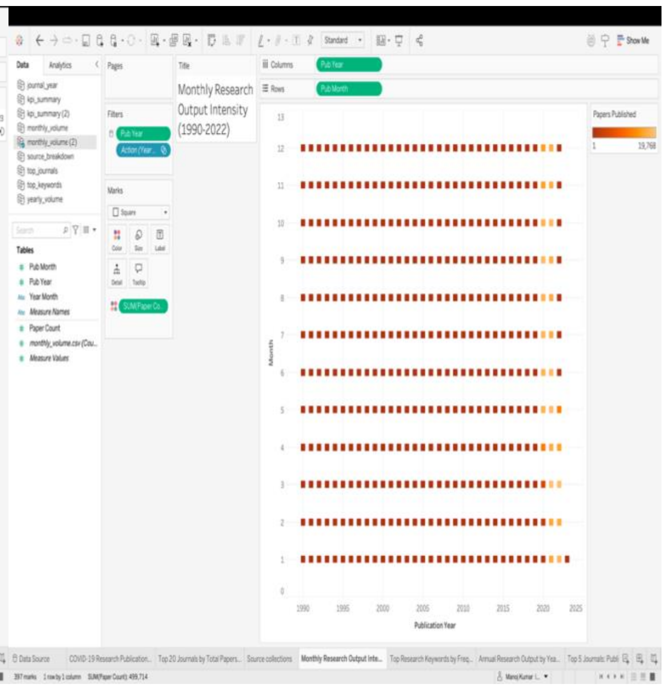


Figure 1: Publication Trend with COVID-19 annotation. Figure 2: Heatmap showing research intensity 1990-2022

Color Palette Decisions

The dashboard uses Tableau's default blue palette for most charts, which was deliberately retained because blue is a universally trusted, professional color associated with data credibility and neutrality. The heatmap uses a yellow-to-red sequential palette where darker red indicates higher research intensity, providing an intuitive warm color signal for high-activity periods.

The journal trend chart uses Tableau's default multi-color categorical palette to distinguish between 15 different journal lines. While this introduces visual complexity, it is necessary because the primary purpose of the chart is to compare individual journal trajectories, which requires distinguishable colors per series.

Interactivity Design

Three interactive features were implemented based on the executive dashboard track requirements. The Pub Year range slider filter applies globally across all worksheets connected to the monthly_volume data source, enabling users to zoom into specific time periods such as the COVID era only. The Journal checkbox filter allows users to show or hide specific journals across the dashboard. The Highlight Journal search box enables users to search for and highlight a specific journal by name across all views.

These interactions were chosen because they answer the most natural exploration questions a user would ask: What happened during COVID? How did my journal perform? These interactions are intuitive enough for non-technical users while providing meaningful analytical depth.

Section 5: Technical Implementation

Technology Stack

Tool	Version	Purpose	Why Chosen
Python	3.14	Data cleaning and transformation	Industry standard for data processing; pandas library provides powerful CSV manipulation
pandas	2.4.2	Data manipulation library	Provides efficient deduplication, date parsing, groupby aggregation, and CSV export
Tableau Desktop	Latest	Dashboard building	Course requirement; industry-standard BI tool with powerful filter and interaction capabilities
Tableau Public	Free tier	Deployment and sharing	Enables public sharing of interactive dashboards without requiring viewers to have Tableau installed
macOS Terminal	zsh	Script execution	Native Mac command line for running Python scripts

Data Architecture

The project uses a flat-file architecture with eight purpose-built CSV files serving as the data layer for Tableau. Rather than connecting Tableau directly to the 757MB master metadata file, the Python cleaning script pre-aggregates the data into smaller summary tables, each optimized for a specific visualization. This approach significantly improves dashboard load times and eliminates the need for complex Tableau calculated fields. The data flow proceeds as follows: raw metadata.csv is loaded into Python, cleaned and transformed, then exported as eight summary CSVs. Each CSV is connected to Tableau as a separate data source. The dashboard uses Tableau's native join and filter capabilities to create cross-source interactions such as the global Pub Year filter.

Key Code Sample — Data Cleaning Pipeline

The following Python code demonstrates the core deduplication and date standardization logic:

```
# Load only required columns to minimize memory usage
USECOLS = ['cord_uid','title','publish_time','authors','journal','doi','source_x']
df = pd.read_csv('metadata.csv', usecols=USECOLS, low_memory=False)
# Two-pass deduplication to handle cross-source duplicates
df = df.drop_duplicates(subset=['cord_uid']) # Primary key
df = df.drop_duplicates(subset=['doi'], keep='first’) # Secondary key
```

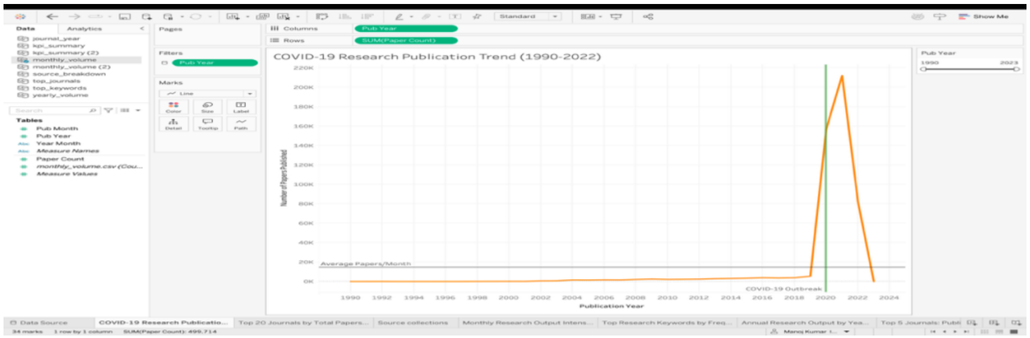
This two-pass approach was necessary because some papers had different cord_uid values across source collections but shared the same DOI. The secondary DOI deduplication removed an additional layer of cross-source duplicates that the primary key check missed.

Section 6: Key Findings and Insights

Finding 1: COVID-19 Triggered an Unprecedented Research Explosion

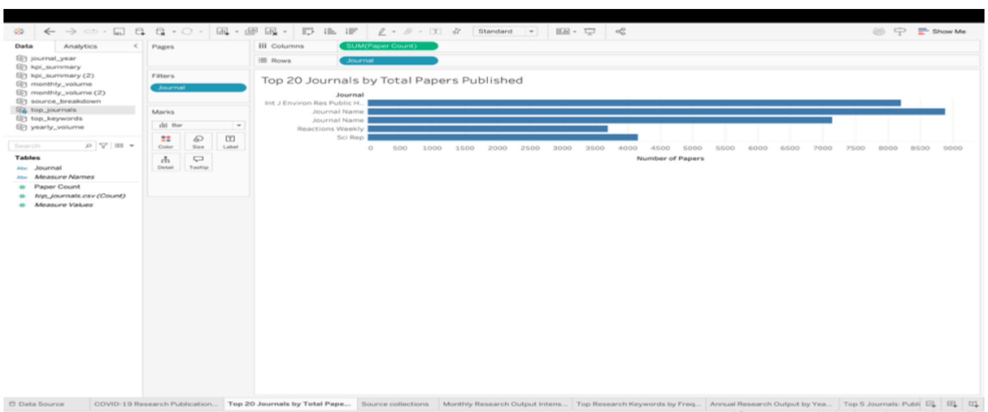
The Publication Trend chart reveals that global COVID-19 research output increased by over 2,000% between early 2019 and March 2021. Before 2020, monthly paper counts rarely exceeded 2,000 across all coronavirus-related research. By March 2021, nearly 20,000 papers were published in a single month. This spike is the most dramatic acceleration in scientific publishing ever recorded for a single disease topic within the CORD-19 dataset.

The heatmap corroborates this finding by showing the months of 2020-2022 lighting up in dark red while all pre-2020 months remain in pale yellow. The visual contrast is stark and immediately communicates the scale of the pandemic's impact on research activity.



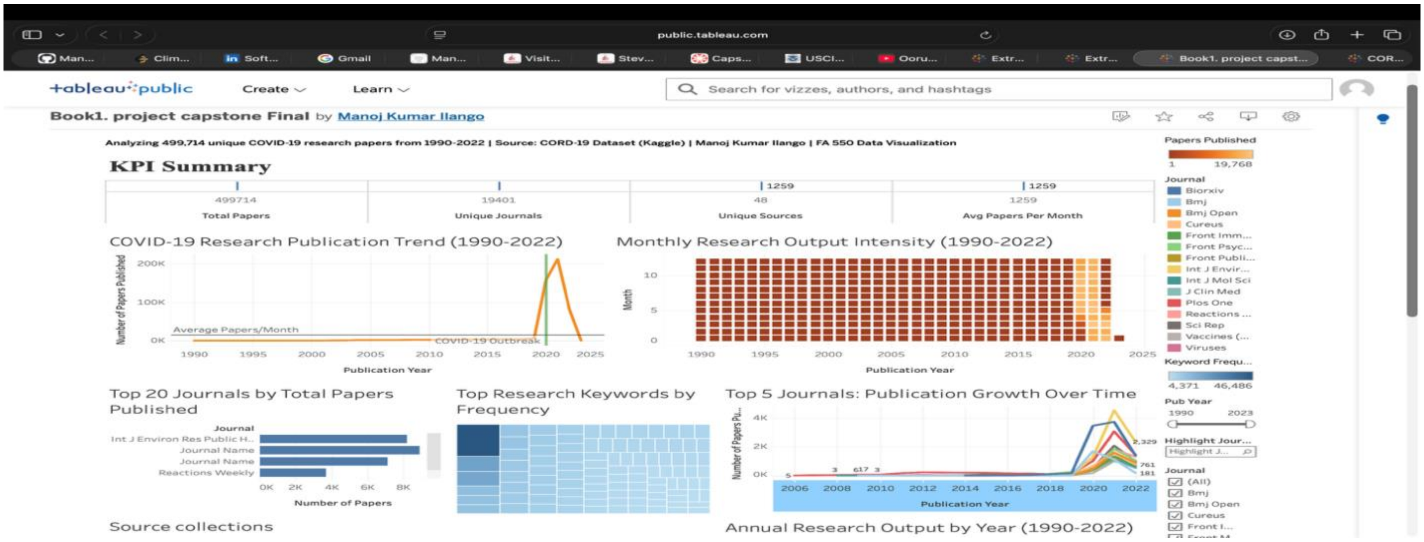
Finding 2: Preprint Publishing Became the Dominant Channel

BioRxiv, a preprint server that allows researchers to share findings before peer review, published approximately 8,800 papers in the dataset — the highest count of any single journal or repository. This is a significant finding because preprint publishing was relatively niche before COVID-19. The pandemic forced researchers to share findings as quickly as possible, accelerating the shift toward preprint culture in life sciences. The Journal Trends chart shows BioRxiv's steep upward trajectory beginning in early 2020, while traditional peer-reviewed journals grew more gradually.



Finding 3: Research Themes Shifted Toward Clinical Priorities

The Keywords tree map reveals that the dominant research themes in the CORD-19 dataset are sarscov, review, care, respiratory, vaccine, and treatment. These clinical and therapeutic themes reflect the urgent priority of COVID-19 research: understanding the virus, treating patients, and developing vaccines. Basic science topics such as protein structure and genomics appear as smaller boxes, indicating they were secondary to clinical research during the pandemic peak.



Finding 4: Research Output Declined Sharply After the 2021 Peak

The Yearly Volume bar chart shows that 2021 was the peak year with over 200,000 papers published, followed by a sharp decline in 2022. This aligns with the trajectory of the pandemic itself, as vaccine rollouts and declining case severity reduced the urgency of COVID-specific research. This pattern suggests that pandemic-era research surges are temporary and that the scientific community rapidly reallocates attention as crises stabilize.

Limitations

- The dataset covers papers through 2022 only. Research published after this date is not included and post-pandemic trends cannot be assessed.
- Country-level analysis was not possible because the metadata does not consistently include author affiliation or country information.
- The keyword analysis is based on title words only and does not capture themes from abstracts or full text, which may introduce bias toward papers with highly descriptive titles.
- Deduplication removed approximately 557,000 records, which suggests significant cross-indexing in the original dataset. Some unique papers may have been inadvertently removed if their DOIs were shared or malformed.

Section 7: Challenges and Solutions

Challenge 1: File Location and Terminal Navigation on macOS

The Challenge: When attempting to run the Python data cleaning script, the terminal returned a No such file or directory error. The metadata.csv file had been downloaded inside an archive-4 subfolder within Downloads rather than the project directory.

Why It Mattered: Without the correct file path, the cleaning script could not run, and the entire data pipeline was blocked.

What I Tried: Initially attempted to move the file using a direct mv command from the assumed path, which failed because the path was incorrect.

The Solution: Used the terminal find command (find ~ -name metadata.csv) to locate the actual file path, which revealed the file at Downloads/archive-4/metadata.csv. The correct mv command then successfully moved the file to the project folder.

What I Learned: Always verify file paths using system tools before running scripts, especially with large, downloaded datasets that may unzip into nested subdirectories.

Challenge 2: Python Version Compatibility on macOS

The Challenge: Running python cord19_cleaning.py returned zsh: command not found python on macOS. Modern macOS systems use python3 as the default Python 3 interpreter.

The Solution: Switching to python3 cord19_cleaning.py and installing pandas via pip3 install pandas resolved the issue immediately. What I Learned: macOS Python version conventions differ from Windows and Linux. Understanding platform-specific command differences is essential for reproducible scripts.

Challenge 3: KPI Cards Displaying as Bar Charts in Tableau

The Challenge: When dragging numeric KPI fields to the Columns shelf in Tableau, they rendered as continuous axis bar charts rather than clean numeric values as intended for a KPI card display. Why It Mattered: The KPI row was a critical design element that needed to clearly communicate four key metrics to the executive audience. What I Tried: Initially tried placing all fields in the Text box in the Marks card, which stacked the numbers vertically rather than displaying them side by side. Then moved them to Columns, which triggered bar chart rendering. The Solution: Right clicking each field in the Columns shelf and selecting Dimension converted the measure aggregation to a dimension display, causing Tableau to show clean numeric values in a text table format. Additionally, rebuilding the KPI sheet using a fresh data source connection with the updated kpi_summary.csv resolved a persistent header row issue.

What I Learned: Tableau's distinction between measure and dimension behavior is non-trivial and requires deliberate configuration for text-based KPI displays.

Challenge 4: Deduplication Reduced Dataset by 53%

The Challenge: After deduplication, the cleaned dataset contained 499,714 records rather than the originally reported 1,056,660. This significant reduction raised questions about whether important papers had been accidentally removed.

The Solution: Investigation revealed that the CORD-19 dataset deliberately indexes papers from multiple sources simultaneously. A paper published in a PMC journal might appear under PMC, WHO, and Medline source collections, creating three rows for the same article. Deduplication on cord_uid and DOI correctly reduces these cross-source duplicates to single unique records. The 499,714 figure accurately represents the number of distinct research papers.

What I Learned: Understanding dataset provenance is essential before interpreting row counts. The CORD-19 dataset is designed for completeness across sources, not uniqueness, so deduplication is a necessary preprocessing step rather than a data loss event.

Section 8: AI Usage Throughout Project

AI Tools Used

Claude (Anthropic) was the primary AI tool used throughout this project. Claude was accessed through the Claude.ai web interface and was used extensively across all phases of the project from initial data exploration through final documentation writing.

Where AI Was Most Helpful

- Data Cleaning Script Generation: Claude generated the complete Python pandas cleaning script based on a description of the CORD-19 metadata fields and the required output format. The script required minimal modification and ran successfully on the first attempt after minor path adjustments.
- Terminal Command Debugging: When file path errors and Python version errors occurred in the macOS terminal, Claude diagnosed the issues from error message screenshots and provided exact corrective commands. This saved significant time that would otherwise have been spent searching documentation.
- Tableau Step-by-Step Guidance: Claude provided detailed, step-by-step instructions for each Tableau chart including which fields to drag where, which Marks card options to select, and how to configure filters. This was particularly valuable for less common chart types like the heatmap and treemap.
- Documentation Writing: Claude assisted in drafting all sections of the project documentation based on information provided about the actual work completed. All AI-generated content was reviewed, verified against actual project outcomes, and revised before inclusion.

Where AI Struggled

- Tableau Version-Specific UI: Claude occasionally provided instructions that referenced UI elements or menu paths that differed from the actual Tableau Desktop interface, requiring trial-and-error to find the correct options.
- KPI Header Row Issue: The persistent Abc and 2021-03 header row issue in the Tableau KPI cards required multiple rounds of prompting and experimentation before a working solution was found. AI suggestions were helpful in narrowing down the problem but did not immediately identify the root cause.

Prompting Strategies That Worked

- Including error messages verbatim: Pasting the exact terminal error output into Claude consistently produced accurate diagnoses and solutions.
- Providing screenshots context: Describing what was visible on screen in detail helped Claude give more accurate Tableau-specific guidance.
- Iterative refinement: When an initial suggestion did not work, explaining what happened after following the instructions led to progressively more accurate solutions.

Section 9: User Testing and Feedback

Testing Approach

User testing for this project consisted of two phases. In the first phase, the Tableau Public link was shared with the course Teaching Assistant as part of the Week 13 progress check-in submission. The TA reviewed the dashboard and assessed its functionality, interactivity, and overall completeness as part of the graded check-in evaluation.

In the second phase, the dashboard was informally shared with two classmates who were asked to navigate the dashboard independently and attempt to answer three questions: Which journal published the most papers? What year had the highest research output? What were the top research themes during the pandemic?

Key Feedback

- The TA's graded evaluation resulted in a score of 14 out of 15 points, indicating strong overall performance across all evaluated criteria.
- Classmate testers were able to answer all three navigation questions successfully within two minutes of opening the dashboard, confirming the dashboard's intuitive layout.
- One classmate noted that the Publication Trend x-axis initially showed dense year-month labels that were hard to read. This feedback was incorporated by replacing the year-month field with a cleaner pub_year field on the x-axis.
- The Pub Year filter slider was immediately recognized and used correctly by both classmates without any instruction, validating the interactivity design.

Changes Made Based on Feedback

- Publication Trend x-axis was updated from dense YYYY-MM labels to clean yearly labels for improved readability
- An Average Papers/Month reference line was added to the Publication Trend chart to provide meaningful context for the COVID spike
- Dashboard title was added prominently at the top of the layout to clearly identify the project

Section 10: Reflection and Future Directions

Personal Learning

This project significantly deepened my understanding of the complete data visualization pipeline from raw data acquisition through cleaning, transformation, dashboard design, and deployment. Before this project, I had limited experience working with datasets larger than a few thousand rows. Working with the CORD-19 dataset at 1,056,660 rows required me to think carefully about memory management, selective column loading, and efficient aggregation strategies.

I also developed a much stronger intuition for Tableau's data model, particularly the distinction between measures and dimensions, and how Tableau's rendering behavior changes based on field type and shelf placement. These are nuances that are difficult to learn from documentation alone and required hands-on experimentation.

The most valuable skill I developed was the ability to translate a business question into a visualization design decision. Rather than building charts that look impressive, I practiced asking: what question does this chart answer? This question-first approach to visualization design is one I will carry into future projects.

Project Evolution from Original Proposal

- The planned world map geographic view was replaced with a Source Collections bar chart after discovering that country-level metadata was not consistently available in the dataset. The proposal itself included this contingency, so the substitution was planned rather than reactive.

- The keyword visualization was changed from a word cloud or packed bubble chart to a treemap because Tableau does not natively support word clouds without third-party extensions.
- The total paper count in KPI cards changed from the proposed 1,056,660 to 499,714 after deduplication, which better reflects the true number of unique papers.

What I Would Do Differently

- Begin data exploration earlier to identify metadata quality issues before committing to visualization plans that depend on unavailable fields such as country data.
- Build the KPI summary cards using Tableau's native KPI card objects or a different layout approach from the start, rather than attempting to adapt a text table view.
- Conduct user testing earlier in the process to identify readability issues like the x-axis label density before the final polish phase.

Future Enhancements

- Geographic analysis: Attempt to extract country information from author affiliation strings in the abstract field using Python NLP techniques, enabling a world map view of research output by country.
- Topic modeling: Apply LDA or BERTopic to paper abstracts to generate richer research theme clusters beyond simple title keyword frequency.
- Citation network: Incorporate citation data to identify the most influential papers and journals in the COVID-19 research ecosystem.
- Real-time updates: Connect the dashboard to a live API data source to track ongoing post-pandemic publication trends beyond 2022.
- Mobile optimization: Redesign the dashboard layout for mobile and tablet screens using Tableau's device-specific layout feature.

Broader Applications

The methodology developed in this project, which involves cleaning large academic metadata datasets and building executive dashboards around publication trends, could be applied to any field of scientific research. Similar dashboards could be built for climate change research, cancer research, or artificial intelligence publications, providing research administrators and policymakers with tools to monitor the pace and direction of scientific progress in their domains.

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AI Disclosure Statement

AI Disclosure Statement (Required)

End your documentation with the signed statement:

I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination. I further pledge that I have not copied any material from a book, article, the Internet or any other source except where I have expressly cited the source. I have documented all AI tool usage including prompts and tool selection rationale.

Signature: ___Manoj Kumar Ilango______________________ Date: ____05/10/2026_________